

Preliminaries

C^k functions: $C^k(\Omega) = \{u: \Omega \rightarrow \mathbb{R} \mid u \text{ and all its partial derivatives up to and including order } k \text{ are in } C^0(\Omega)\}$

The sets C^k are function spaces.

$(C^k(\bar{\Omega}), \|\cdot\|_{C^k(\bar{\Omega})})$ is a Banach space, with norm

$$\|f\|_{C^k(\Omega)} = \sum_{|\alpha| \leq k} \sup_{x \in \Omega} |D^\alpha f(x)|$$

Convex:

One-dim case:

$$f \in C^1[a, b] \Rightarrow f(x) \geq f(y) + f'(y)(x-y)$$

$$f \in C^2[a, b] \Rightarrow f''(x) \geq 0$$

Higher dims:

$$\bullet f(\vec{x}) \geq f(\vec{y}) + \nabla f(\vec{y})(\vec{x} - \vec{y})$$

$$\bullet \text{ Hessian matrix } \nabla^2 f(\vec{x}) = \left\{ \left(\frac{\partial^2 f}{\partial x_i \partial x_j} \right) \right\}_{i,j=1}^d \text{ is positive semi-definite}$$

Jensen's Inequality

$u: [a, b] \rightarrow \mathbb{R}$ is a function. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be convex. Then

$$f\left(\frac{1}{b-a} \int_a^b u(x) dx\right) \leq \frac{1}{b-a} \int_a^b f(u(x)) dx$$

L^p norm:

$$\|u\|_{L^p(\Omega)} = \left(\int_{\Omega} |u(x)|^p dx \right)^{\frac{1}{p}}$$

$$\|u\|_{L^\infty(\Omega)} = \operatorname{ess\,sup}_{x \in \Omega} |u(x)|$$

Hölder Inequality

Let $p, q \in [1, +\infty]$, and $\frac{1}{p} + \frac{1}{q} = 1$

If $u \in L^p(\Omega)$, $v \in L^q(\Omega)$, then $uv \in L^1(\Omega)$ with

$$\int_{\Omega} |u(x)v(x)| dx \leq \|u\|_{L^p(\Omega)} \|v\|_{L^q(\Omega)}$$

Minkowski Inequality

$$\|u+v\|_{L^p(\Omega)} \leq \|u\|_{L^p(\Omega)} + \|v\|_{L^p(\Omega)}$$

Density of smooth functions

For $p \in [1, +\infty)$, the functions in $C_c^\infty(\Omega)$ are dense in $L^p(\Omega)$

Sequences and convergence of functions in $L^p(\Omega)$

Definition: **Strong convergence.**

A sequence $\{u_n\}$ of functions in $L^p(\Omega)$ strongly converge to $u \in L^p(\Omega)$ if

$$\lim_{n \rightarrow +\infty} \|u_n - u\|_{L^p(\Omega)} = 0$$

Notation: $u_n \rightarrow u$ in $L^p(\Omega)$

Definition: **Weak convergence**

For $p \in [1, +\infty)$, a sequence $\{u_n\}$ in $L^p(\Omega)$ weakly converge to $u \in L^p(\Omega)$ if

$$\lim_{n \rightarrow +\infty} \int_{\Omega} (u_n(x) - u(x)) \varphi(x) dx = 0 \quad \text{for any } \varphi(x) \in L^q(\Omega) \text{ s.t. } \frac{1}{p} + \frac{1}{q} = 1$$

Notation: $u_n \rightharpoonup u$ in $L^p(\Omega)$

If $p = +\infty$, a sequence $\{u_n\}$ of functions in $L^\infty(\Omega)$

weak-* converges to $u \in L^\infty(\Omega)$ if

$$\lim_{n \rightarrow +\infty} \int_{\Omega} (u_n(x) - u(x)) \varphi(x) dx = 0 \quad \forall \varphi \in L^1(\Omega)$$

Notation: $u_n \xrightarrow{*} u$ in $L^\infty(\Omega)$

Remark:

- 1) Whether weak or strong, the limit is unique.
- 2) Strong convergence $\Rightarrow$ Weak convergence

Theorem: for $\Omega \subseteq \mathbb{R}^d$ a bounded region, the following hold:

1) $u_n \xrightarrow{*} u$ in $L^\infty(\Omega) \Rightarrow u_n \rightarrow u$ in $L^p(\Omega)$ for $p \in [1, +\infty)$

2) $u_n \rightarrow u$ in $L^p(\Omega) \Rightarrow \|u_n\|_{L^p(\Omega)} \rightarrow \|u\|_{L^p(\Omega)}$ for $p \in [1, +\infty)$

3) If $1 \leq p < +\infty$, $u_n \rightarrow u$ in $L^p(\Omega)$
 $u_n \not\rightarrow u$ in $L^\infty(\Omega)$

Then $\exists M > 0$ s.t.

$$\|u_n\|_{L^p(\Omega)} \leq M \quad \forall n$$

Furthermore, $\|u\|_{L^p(\Omega)} \leq \liminf_{n \rightarrow +\infty} \|u_n\|_{L^p(\Omega)}$

4) For $p \in [1, +\infty]$, if $\exists M$ s.t. $\|u_n\|_{L^p(\Omega)} \leq M \quad \forall n$, then $\exists$ a subsequence

$\{u_{n_k}\} \rightarrow u$ in $L^p(\Omega)$ for $1 \leq p < +\infty$

$\{u_{n_k}\} \not\rightarrow u$ in $L^\infty(\Omega)$

Comment: Bolzano-Weierstrass theorem for L^p spaces works for weak convergences.

5) For $p \in [1, +\infty]$, and $u_n \rightarrow u$ in $L^p(\Omega)$, $\exists$ a subsequence $\{u_{n_k}\}$ s.t. $u_{n_k}(x) \rightarrow u(x)$ for a.e. $x \in \Omega$.

Riemann-Lebesgue Lemma

Let $p \in [1, +\infty]$, $u \in L^p[a, b]$, extend u periodically from (a, b) to $\mathbb{R}$.

Define $u_n(x) = u(nx)$. $\bar{u} = \frac{1}{b-a} \int_a^b u(x) dx$

Then $u_n \rightharpoonup \bar{u}$ in $L^p(a, b)$ for $p \in [1, +\infty)$
 $u_n \not\rightarrow \bar{u}$ in $L^\infty(a, b)$

Dimension 2: $\Omega = (a, b) \times (c, d)$, $u \in L^p(\Omega)$; extend u periodically to $\mathbb{R}^2$

$$\bar{u} = \frac{1}{|\Omega|} \iint_{\Omega} u(\vec{x}) d\vec{x}$$

Then $u_n(x) = u(nx)$ converges weakly in $L^p(\Omega)$. weak- $*$ in $L^\infty(\Omega)$ to $\bar{u}$.

Definition: $\Omega \subseteq \mathbb{R}^d$, $u \in L^p_{loc}(\Omega)$ if $u \in L^p(K)$ for every compact $K \subseteq \Omega$.

Fundamental Lemma of the Calculus of Variations, Revisited

Theorem: Let $\Omega \subseteq \mathbb{R}^d$, $u \in L^1_{loc}(\Omega)$ satisfy $\int_{\Omega} u(x) \varphi(x) dx = 0$

$\forall \varphi \in C_c^\infty(\Omega)$; then $u(x) = 0$ a.e. in Ω .

Corollary: $\int_{\Omega} u(x) \varphi(x) dx = 0$ with $\int_{\Omega} \varphi(x) dx = 0 \Rightarrow u(x)$ is constant a.e.

Sobolev Spaces

Definition: **Weak derivative**

For $u \in L^1_{loc}(\Omega)$, $v \in L^1_{loc}(\Omega)$ is its weak derivative w.r.t. x_i if $\int_{\Omega} v(x) \varphi(x) dx = - \int_{\Omega} u(x) \frac{\partial \varphi}{\partial x_i}(x) dx \forall \varphi \in C_c^\infty(\Omega)$.

Definition: **Sobolev spaces**

1) $W^{1,p}(\Omega)$ = the set of functions $u: \Omega \rightarrow \mathbb{R}$ s.t.

$u \in L^p(\Omega)$ and $\frac{\partial u}{\partial x_i} \in L^p(\Omega)$ for $i=1, 2, \dots, d$
weak derivative

$W^{1,p}(\Omega)$ is a Banach space with norm

$$\|u\|_{W^{1,p}(\Omega)} = \left(\|u\|_{L^p(\Omega)}^p + \|\nabla u\|_{L^p(\Omega)}^p \right)^{\frac{1}{p}} \text{ for } p \in [1, +\infty)$$

$$\|u\|_{W^{1,\infty}(\Omega)} = \max \{ \|u\|_{L^\infty(\Omega)}, \|\nabla u\|_{L^\infty(\Omega)} \}, \quad p = +\infty$$

2) For $p \in [1, +\infty)$, $W_0^{1,p}(\Omega) = \{ u \in W^{1,p}(\Omega) : \exists \text{ a sequence } \{u_n\} \text{ in } C_0^\infty(\Omega) \text{ that converges to } u \text{ in } W^{1,p}(\Omega) \}$

Informal definition $W_0^{1,p}(\Omega) = \{ u \in W^{1,p}(\Omega) : u = 0 \text{ on } \partial\Omega \}$

$$\text{For } p = +\infty, \quad W_0^{1,\infty}(\Omega) = W^{1,\infty}(\Omega) \cap W_0^{1,1}(\Omega)$$

3) For $k \in \mathbb{N}$, $W^{k,p}(\Omega)$ = the set of functions $u: \Omega \rightarrow \mathbb{R}$ s.t. $u \in L^p(\Omega)$ and all weak derivatives up to and including order k are in $L^p(\Omega)$.

$$\|u\|_{W^{k,p}(\Omega)} = \left(\|u\|_{W^{k-1,p}(\Omega)}^p + \|\nabla u\|_{W^{k-1,p}(\Omega)}^p \right)^{\frac{1}{p}}$$

$$W_0^{k,p}(\Omega) = \{ u \in W^{k,p}(\Omega) : u = 0 \text{ on } \partial\Omega \}$$

Convention: $W^{0,p}(\Omega) = L^p(\Omega)$

Remarks:

1) If Ω is bounded,

$$C^1(\bar{\Omega}) \subseteq W^{1,\infty}(\Omega) \subset W^{1,p}(\Omega) \subset L^p(\Omega)$$

2) For $p = +\infty$, $\partial\Omega$ smooth, $W^{1,\infty}(\Omega) = \{ u \in C^0(\Omega) : u \text{ is Lipschitz} \}$

3) For $p \in [1, +\infty)$, if $\exists M$ s.t. $\|u_n\|_{W^{1,p}(\Omega)} \leq M$, then $\exists$ a subsequence $\{u_{n_k}\}$ and $u \in W^{1,p}(\Omega)$ s.t. $u_{n_k} \rightarrow u$ in $W^{1,p}(\Omega)$

True for $\xrightarrow{*}$ for $p = +\infty$.

Theorem: Weak convergence in $W^{1,p}$ implies strong convergence in L^p

$\Omega \subseteq \mathbb{R}^d$ with regular boundary. $p \in [1, +\infty)$. If $u_n \rightharpoonup u$ in $W^{1,p}(\Omega)$, then $u_n \rightarrow u$ in $L^p(\Omega)$.

For $p = +\infty$, $u_n \xrightarrow{*} u$ in $W^{1,\infty}(\Omega) \Rightarrow u_n \rightarrow u$ in $L^\infty(\Omega)$.

Theorem: Poincaré Inequality

Let $\Omega \subseteq \mathbb{R}^d$ be a bounded region. $p \in [1, +\infty]$. Then $\exists C > 0$ depending only on Ω and p s.t.

$$\|u\|_{L^p(\Omega)} \leq C \|\nabla u\|_{L^p(\Omega)} \quad \forall u \in W_0^{1,p}(\Omega)$$

Equivalently,

$$\|u\|_{W^{1,p}(\Omega)} \leq \tilde{C} \|\nabla u\|_{L^p(\Omega)}$$

Another form of the Poincaré Inequality:

$\Omega \subseteq \mathbb{R}^d$ bounded, $p \in [1, +\infty]$, $u \in W^{1,p}(\Omega)$. Suppose $\exists g \in W^{1,p}(\Omega)$, s.t. $u(x) - g(x) \in W_0^{1,p}(\Omega)$.

Then there exist $M_1, M_2 > 0$ depending only on p and Ω , s.t.

$$M_1 \|u\|_{W^{1,p}(\Omega)} - M_2 \|g\|_{W^{1,p}(\Omega)} \leq \|\nabla u\|_{L^p(\Omega)}$$

Remark: This theorem is false for $u \in W^{1,p}(\Omega)$, that $u \neq 0$ on $\partial\Omega$

Proof omitted.